Wk11L1: Impedance & sinusoidal analysis

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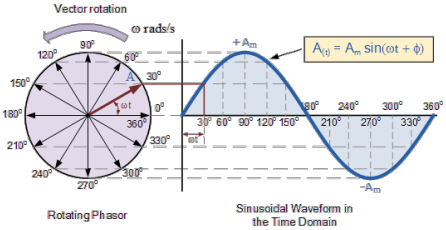

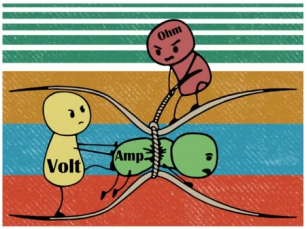

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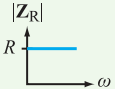
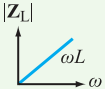
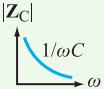
Table 7-3: Time-domain sinusoidal functions $x(t)$ and their cosine-reference phasor-domain counterparts $\mathbf{X}$, where $x(t) = \Re e\ [\mathbf{X}e^{j\omega t}]$.

$x(t)$		$\mathbf{X}$
$A\cos\omega t$	$\longleftrightarrow$	A
$A\cos(\omega t+\phi)$	$\longleftrightarrow$	$Ae^{j\phi}$
$-A\cos(\omega t+\phi)$	$\longleftrightarrow$	$Ae^{j(\phi\pm\pi)}$
$A\sin\omega t$	$\longleftrightarrow$	$Ae^{-j\pi/2} = -jA$
$A\sin(\omega t+\phi)$	$\longleftrightarrow$	$Ae^{j(\phi-\pi/2)}$
$-A\sin(\omega t+\phi)$	$\longleftrightarrow$	$Ae^{j(\phi+\pi/2)}$
$\frac{d}{dt}(x(t))$	$\longleftrightarrow$	$j\omega\mathbf{X}$
$\frac{d}{dt}[A\cos(\omega t+\phi)]$	$\longleftrightarrow$	$j\omega Ae^{j\phi}$
$\int x(t)\,dt$	$\longleftrightarrow$	$\frac{1}{j\omega}\mathbf{X}$
$\int A\cos(\omega t+\phi)\,dt$	$\longleftrightarrow$	$\frac{1}{j\omega}Ae^{j\phi}$

Table 7-4: Summary of v–i properties for R, L, and C.

Property	R	L	C
v–i	$v = Ri$	$v = L\dfrac{di}{dt}$	$i = C\dfrac{dv}{dt}$
V–I	$\mathbf{V} = R\mathbf{I}$	$\mathbf{V} = j\omega L\mathbf{I}$	$\mathbf{V} = \dfrac{\mathbf{I}}{j\omega C}$
Z	R	$j\omega L$	$\dfrac{1}{j\omega C}$
dc equivalent	R	Short circuit	Open circuit
High-frequency equivalent	R	Open circuit	Short circuit
Frequency response			

Phasor .gif
https://en.wikipedia.org/wiki/Phasor

Adding impedances

Same as resistances and inductors

$$Z_{eq,--} = \sum_{k=1}^{n} Z_k \tag{1}$$

$$Z_{eq,||} = \left(\sum_{k=1}^{n} \frac{1}{Z_k} \right)^{-1} \tag{2}$$

Voltage division

Same as resistances

$$\widetilde{V}_Z = \frac{Z}{Z_{eq}} \widetilde{V}_{Z_{eq}} \tag{3}$$

Thévenin equivalents

Same as resistances

Find open-circuit voltage phasor $\Rightarrow \widetilde{V}_{oc}$
Find short-circuit current phasor $\Rightarrow \widetilde{I}_{sc}$

$$\widetilde{V}_{Th} = \widetilde{V}_{oc} \tag{4}$$

$$Z_{Th} = \frac{\widetilde{V}_{oc}}{\widetilde{I}_{sc}} \tag{5}$$

Nodal and mesh analysis

Same as resistances

- Kirchhoff's voltage law (KVL) applies to voltage phasors
- Kirchhoff's current law (KCL) applies to current phasors
- Similar to Ohm's law you may use $\widetilde{V} = \widetilde{I}Z$
- All other nodal and mesh analysis techniques still apply

Op-amps

Same as resistances

May still use ideal op-amp model (use caution since this model is less applicable at high frequency in real life).

$$\widetilde{V}_p = \widetilde{V}_n \tag{6}$$

$$\widetilde{I}_p = \widetilde{I}_n = 0 \tag{7}$$

Reminders

- HW 8 is posted